

# System Configuration

# Objectives

After completing this lesson, you should be able to describe:

- System date and time configuration
- NTP and chrony configuration
- The `/etc/sysconfig` directory
- The `proc` file system
- The `sysctl` utility
- The `sysfs` file system
- The `lshw` and `lshw-qui` utilities



# Configuring System Date and Time During Installation

DATE & TIME ORACLE LINUX 7.5 INSTALLATION

Done us Help!

Region: Americas City: Denver Network Time OFF



09:48 PM ☒ 24-hour ☐ AM/PM

04 / 18 / 2018

 You need to set up networking first if you want to use NTP.

# Configuring System Date and Time from the Command Line

- Oracle Linux has three command-line utilities to configure the system date and time:
  - `date`
  - `hwclock`
  - `timedatectl`
- Run the `date` command to display or set the system date and time.
  - Can also be used to display dates in the past or future
- Run the `hwclock` command to display or set the hardware clock date and time.
  - Can sync hardware clock with system time
  - Can sync system time with hardware clock



# Using the `timedatectl` Utility

- The `timedatectl` utility is part of the `systemd` system and service manager.
- To display local, universal, RTC time, time zone, NTP configuration, and DST information:

```
# timedatectl
```

- Use the following syntax to change the date and time:

```
# timedatectl set-time <YYYY-MM-DD>
```

```
# timedatectl set-time <HH:MM:SS>
```

- Use the following syntax to change the time zone:

```
# timedatectl set-timezone <time_zone>
```

- To enable clock synchronization over NTP:

```
# timedatectl set-ntp yes
```

# Using Network Time Protocol

- Synchronize your computer's time with another system.
  - NTP servers are arranged in layers, or strata, from 1 to 15.
  - Specify NTP servers in `/etc/ntp.conf`.

- Start the `ntpd` service as follows:

```
# systemctl start ntpd
```

- Use the `ntpq` command to query NTP operations and performance.
  - To view which NTP server your system is synchronized with:

```
# ntpq -p
```

- Use the `ntpstat` command to view network time synchronization status:

```
# ntpstat  
synchronized to NTP server (10.150.36.208) at stratum 5  
...
```

# Configuring NTP by Using Chrony

- Chrony provides another implementation of NTP.
- Chrony is designed for systems that are often powered down or disconnected from the network.
- The main configuration file is `/etc/chrony.conf`.
  - Parameters are similar to those in the `/etc/ntp.conf` file.
- `chronyd` is the daemon that runs in user space.
- `chronyc` is a command-line program that provides a command prompt and several commands. Examples:
  - `tracking`: Displays system time information
  - `sources`: Displays information about current sources
- The user guide is accessible at <http://chrony.tuxfamily.org/manual.html>.

# The `/etc/sysconfig` Directory

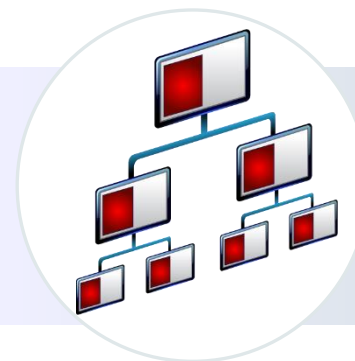
- The `/etc/sysconfig` directory contains a hierarchy of system configuration files.
- For complete information about these files, see `/usr/share/doc/initscripts*/sysconfig.txt`.
- The contents of the directory vary depending on the programs installed.
- Files in the directory describe:
  - Configuration parameters
  - Arguments for respective daemons
  - Custom options for commands
  - System defaults
  - Kernel information
  - Much more





# The `proc` File System

- The `proc` file system is a hierarchy of special files that represent:
  - The current state of the kernel
  - Details of system hardware
  - Running processes
  - System configuration information and interfaces
- It is a virtual file system containing virtual files.
- Use `cat`, `more`, and `less` to view most files.
- Utilities exist to view other files. Example:
  - `lspci`, `free`, `top`
- Some files can be modified to adjust kernel settings:
  - `echo value > /proc/<file or path-to-file>`



# Top-Level Files in /proc

|      |      |      |      |      |     |     |            |             |              |               |
|------|------|------|------|------|-----|-----|------------|-------------|--------------|---------------|
| 1    | 1244 | 1410 | 243  | 3331 | 484 | 545 | 911        | dma         | loadavg      | sys           |
| 10   | 1285 | 1415 | 245  | 3336 | 485 | 547 | 916        | driver      | locks        | sysrq-trigger |
| 103  | 1293 | 1420 | 246  | 3600 | 49  | 578 | 923        | execdomains | mdstat       | sysvipc       |
| 1049 | 1294 | 1426 | 25   | 364  | 5   | 579 | 924        | fb          | meminfo      | thread-self   |
| 1051 | 13   | 1435 | 250  | 3713 | 50  | 582 | 927        | filesystems | misc         | timer_list    |
| 11   | 1303 | 1456 | 2597 | 3771 | 502 | 588 | 928        | fs          | modules      | timer_stats   |
| 1152 | 1316 | 15   | 26   | 3796 | 522 | 590 | 947        | interrupts  | mounts       | tty           |
| 1153 | 1323 | 16   | 2602 | 3797 | 523 | 591 | acpi       | iomem       | mtrr         | uptime        |
| 1186 | 1328 | 17   | 27   | 392  | 525 | 6   | buddyinfo  | ioports     | net          | version       |
| 12   | 1330 | 18   | 270  | 398  | 526 | 600 | bus        | irq         | pagetypeinfo | vmallocinfo   |
| 1202 | 1337 | 2    | 28   | 402  | 527 | 601 | cgroups    | kallmodsyms | partitions   | vmstat        |
| 1209 | 1338 | 20   | 280  | 44   | 528 | 7   | cmdline    | kallsyms    | sched_debug  | xen           |
| 1212 | 1345 | 21   | 281  | 45   | 529 | 70  | consoles   | kcore       | scsi         | zoneinfo      |
| 1215 | 1350 | 22   | 29   | 46   | 530 | 71  | cpuinfo    | keys        | self         |               |
| 1219 | 1360 | 23   | 3    | 47   | 532 | 8   | crypto     | key-users   | slabinfo     |               |
| 1224 | 1395 | 239  | 31   | 48   | 533 | 9   | cyclicinfo | kmsg        | softirqs     |               |
| 1227 | 14   | 24   | 32   | 481  | 542 | 907 | devices    | kpagecount  | stat         |               |
| 1236 | 1400 | 241  | 3311 | 482  | 544 | 910 | diskstats  | kpageflags  | swaps        |               |

# Process Directories in `/proc`

- Process directories have names using process ID (PID) numbers.
- They provide information about processes:
  - `cmdline`: The command issued
  - `cwd`: Symbolic link to the current working directory
  - `environ`: Environment variables
  - `fd`: File descriptors
  - `maps`: Memory maps to executables and library files
  - `root`: Symbolic link to the root directory of the process
  - `status`: Run state and memory usage

# Other Directories in `/proc`

- Directories group information concerning the kernel.
- Examples include:
  - `/proc/bus`: Available buses
  - `/proc/driver`: Drivers in use by the kernel
  - `/proc/fs`: Exported file systems
  - `/proc/net`: Network parameters and statistics
  - `/proc/scsi`: SCSI devices

# The `/proc/sys` Directory

- `/proc/sys` has a dual role.
- Provides information about the system
- Contains files used to enable, disable, or modify kernel features
  - Files with write permission may be used to configure the kernel.
  - Use `echo value > filename` to make changes.
  - Changes are not persistent across reboot.
  - Some files contain Boolean values to turn features on or off.
- Subdirectories in `/proc/sys` include `dev`, `fs`, `kernel`, and `net`.



# The `sysctl` Utility

- The `sysctl` utility is used to assign values to writable files in `/proc/sys`.
- To view kernel settings, enter:

```
# sysctl -a
```

- To assign values, the syntax is:

```
# sysctl -w <kernel parameter>=<value>
```

- Example:

```
# sysctl -w net.ipv4.ip_forward=1
```

- Changes are lost when the system is rebooted.
- To preserve settings, add them to files created in `/etc/sysctl.d`
  - Use `sysctl -p <filename>` to make changes immediate.

# The `sysfs` File System

- Beginning with version 2.6, the kernel also exports information to another virtual file system called `sysfs`.
- `sysfs` is mounted on `/sys`.

```
[root@host03 ~]# ls -l /sys
total 0
drwxr-xr-x.  2 root root 0 Sep 26 08:17 block
drwxr-xr-x. 31 root root 0 Sep 26 08:17 bus
drwxr-xr-x. 57 root root 0 Sep 26 08:17 class
drwxr-xr-x.  4 root root 0 Sep 26 08:17 dev
drwxr-xr-x. 19 root root 0 Sep 26 08:17 devices
drwxr-xr-x.  5 root root 0 Sep 26 08:17 firmware
drwxr-xr-x.  8 root root 0 Sep 26 08:17 fs
drwxr-xr-x.  5 root root 0 Sep 26 08:17 hypervisor
drwxr-xr-x. 10 root root 0 Sep 26 08:17 kernel
drwxr-xr-x. 157 root root 0 Sep 26 08:44 module
drwxr-xr-x.  2 root root 0 Sep 26 08:17 power
```

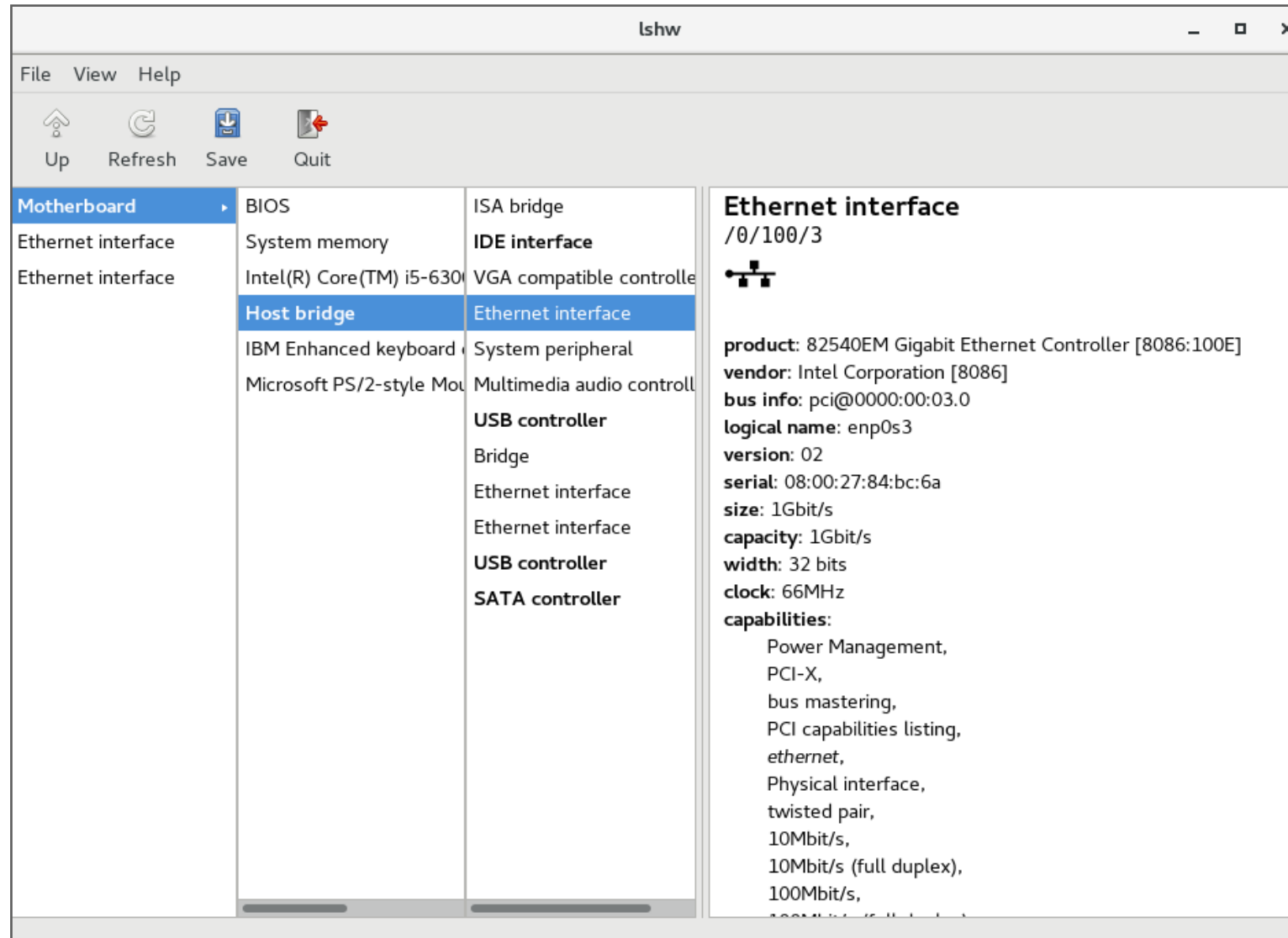
# The lshw Utility

- `lshw` is a command-line tool that provides hardware configuration information.
- Example output for a virtual machine:

```
# lshw
host03.example.com
  description: Computer
  product: HVM domU
  vendor: Xen
  version: 4.1.30VM
  serial: ...
  width: 64 bits
  capabilities: smbios-2.4 dmi-2.4 vsyscall32
  configuration: boot=normal uuid=...
...
```



# The `lshw-gui` Utility: Graphical Hardware Configuration Interface



# Quiz



Which of the following are command-line utilities to configure the system date and time?

- a. date
- b. hwclock
- c. timedatectl
- d. datetimestctl
- e. chrony



# Quiz



Which of the following directories contains files that you can edit to pass arguments to daemons?

- a. `/etc/sysconfig`
- b. `/proc/sys`
- c. `/var/spool`
- d. `/var/log`



# Quiz



Which of the following directories contains files that allow you to enable and disable kernel features in real time?

- a. `/etc/sysconfig`
- b. `/proc/sys`
- c. `/var/spool`
- d. `/var/log`



# Quiz



Which of the following contains process directories named after a program's process ID, each containing information specific to that process?

- a. /etc
- b. /proc/sys
- c. /proc/pid
- d. /proc



# Summary

In this lesson, you should have learned how to describe:

- System date and time configuration
- NTP and chrony configuration
- The `/etc/sysconfig` directory
- The `proc` file system
- The `sysctl` utility
- The `sysfs` file system
- The `lshw` and `lshw-qui` utilities



# Practice 6: Overview

This practice covers the following topics:

- Configuring date and time
- Configuring NTP and chrony
- Exploring the `/etc/sysconfig` directory
- Exploring the `proc` file system
- Exploring the `sysfs` file system
- Using the `sysctl` utility